

Mathias Dick

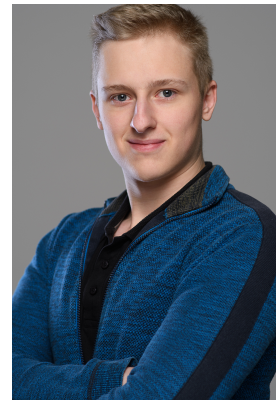
📍 Rastatt, Germany ✉ mathias.dick16@gmail.com

🌐 Portfolio: mathiasd.de 🐙 github.com/MathiasDick 🌐 linkedin.com/in/mathias-dick

Summary

Multidisciplinary engineer combining software development, electronics, and hands-on mechanical expertise. Currently a Dual Student at **Mercedes-Benz AG**, bridging full-stack and desktop application development with automotive engineering. Strong focus on building reliable production-grade tools — from AI-assisted defect ticket generation to industrial network monitoring systems.

🔗 Detailed technical write-ups, architecture diagrams, and photos of my projects are available at mathiasd.de/work



Experience

Dual Student

Oct 2023 – Sep 2026 (Expected)

Mercedes-Benz AG · Rastatt, Germany

AI Ticketing Assistant

Tech: Python · React · Flask · GPT-4o · Whisper · RAG

- Built a hybrid AI pipeline using local Whisper transcription and GPT-4o to automate formal defect ticket generation, reducing manual entry time by 50% for 80+ tickets weekly.
- Implemented RAG-based duplicate detection over vectorized defect history to prevent redundant submissions.
- Deployed as a standalone executable generating import-ready tickets and HTML reports.

Industrial Redundancy Monitor **Tech:** Python · AsyncIO · Flask · React · SNMP · SQLite

- Developed a monitoring system detecting silent redundancy failures across 500+ industrial switches to prevent downtime.
- Engineered an async SNMP scanner with auto-discovery, eliminating manual inventory management.
- Built a virtualized React frontend and a secure, JWT-protected API deployed via IIS.

Freelance Developer

Jan 2024 – Present

Market Monitor

Tech: Python · PostgreSQL · SQLAlchemy · n8n · Linux

- Engineered a production data pipeline scanning 170,000+ products across major e-commerce platforms to enable trend analysis and cross-market price comparison.
- Implemented concurrent workers with process locking and AI-driven data enrichment via n8n, identifying over 50,000 incorrectly listed items.
- Self-hosted on Linux, independently managing deployments, updates, and workflow monitoring.

Selected Projects

Audi Engine Rebuild

Jun 2025

Diagnostics · Precision Metrology

Diagnosed and repaired a catastrophic valve failure on a 2.0 TFSI engine. Applied systematic troubleshooting and factory tolerances for a reliable restoration.

RFID Smart Door Lock

Feb 2024

Arduino · C++ · 3D Printing · CAD

Designed a keyless entry system using RFID and stepper motor control, featuring custom 3D-printed non-destructive mounting hardware.

Education

B.Eng. Embedded Systems

Automotive Engineering Track

DHBW Ravensburg

Campus Friedrichshafen

Oct 2023 – Sep 2026 (Expected)

Dual-track degree in cooperation with Mercedes-Benz AG — alternates between corporate project work and academic semesters.

Skills

Languages & Runtimes

Python, JavaScript / TypeScript, C++ (Embedded), SQL

Frameworks & Libraries

React, Redux Toolkit, Vite, Flask, Waitress, SQLAlchemy, Alembic, Pandas, AsyncIO, PySide6, Selenium

AI / ML

Azure OpenAI (GPT-4o), Whisper (local), RAG / vector embeddings, Prompt engineering

Infrastructure

PostgreSQL, SQLite, n8n, IIS, Windows Server, Linux, Git, PyInstaller

Hardware & Tooling

Arduino / Embedded Systems, SNMP, Fusion 360, Electronics Repair & Diagnostics

Languages

- **German** — Native
- **English** — Fluent
- **Russian** — Intermediate
- **Japanese** — Intermediate (JLPT N2 Expected Juli 2026)